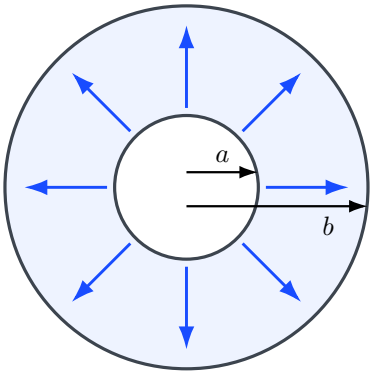
同轴圆柱电容器：径向场决定电势差、能量与内柱外向电压力



$$E_\rho = \frac{\lambda}{2\pi\varepsilon_0\rho}, \quad \Delta V = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{b}{a}$$

$$C = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}, \quad \frac{U}{L} = \frac{\lambda^2}{4\pi\varepsilon_0}\ln\frac{b}{a}$$

$$p(a) = \frac{\varepsilon_0 E(a)^2}{2} = \frac{\lambda^2}{8\pi^2\varepsilon_0 a^2}$$，方向径向向外。